Laplace Transform Tables

Specific Unilateral Laplace Transform Pairs			
Signal	$x(t) = \mathcal{L}^{-1}[X(s)]$		$X(s) = \int_0^\infty x(t)e^{-st}dt$
Unit impulse	$\delta(t)$	$\Longleftrightarrow$	1
Unit step	$u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$	$\Longleftrightarrow$	$\frac{1}{s}$
Exponential	$e^{-\alpha t}\, u(t)$	$\Longleftrightarrow$	$\frac{1}{s+\alpha}$
Power of t	$t^n\, u(t)$	$\Longleftrightarrow$	$\frac{n!}{s^{n+1}}$
Damped power of t	$t^n\, e^{-\alpha t}\, u(t)$	$\Longleftrightarrow$	$\frac{n!}{(s+\alpha)^{n+1}}$
Sine	$\sin(\beta t)\, u(t)$	$\Longleftrightarrow$	$\frac{\beta}{s^2+\beta^2}$
Cosine	$\cos(\beta t)\, u(t)$	$\Longleftrightarrow$	$\frac{s}{s^2+\beta^2}$
Damped sine	$e^{-\alpha t}\, \sin(\beta t)\, u(t)$	$\Longleftrightarrow$	$\frac{\beta}{(s+\alpha)^2+\beta^2}$
Damped cosine	$e^{-\alpha t}\, \cos(\beta t)\, u(t)$	$\Longleftrightarrow$	$\frac{s+\alpha}{(s+\alpha)^2+\beta^2}$
t times damped sine	$t\, e^{-\alpha t}\, \sin(\beta t)\, u(t)$	$\Longleftrightarrow$	$\frac{2\beta\,(s+\alpha)}{((s+\alpha)^2+\beta^2)^2}$
t times damped cosine	$t\, e^{-\alpha t}\, \cos(\beta t)\, u(t)$	$\Longleftrightarrow$	$\frac{(s+\alpha)^2-\beta^2}{((s+\alpha)^2+\beta^2)^2}$

Unilateral Laplace Transform Properties			
Property	$x(t) = \mathcal{L}^{-1}[X(s)]$		$X(s) = \int_0^\infty x(t)e^{-st}dt$
Linearity	$Ax_1(t) + Bx_2(t)$	$\Longleftrightarrow$	$AX_1(s) + BX_2(s)$
Scale change	$x(at)$	$\Longleftrightarrow$	$\frac{1}{a} X\left(\frac{s}{a}\right),\ a > 0$
Time shift	$x(t-T)\, u(t-T)$	$\Longleftrightarrow$	$e^{-sT}\, X(s),\ T > 0$
s-shift	$e^{-\alpha t}\, x(t)$	$\Longleftrightarrow$	$X(s+\alpha)$
Differentiation	$\frac{dx(t)}{dt}$	$\Longleftrightarrow$	$s\, X(s) - x(0^-)$
Times t	$t\, x(t)$	$\Longleftrightarrow$	$\frac{-dX(s)}{ds}$
Integration	$\int_0^t x(\tau)\, d\tau$	$\Longleftrightarrow$	$\frac{X(s)}{s}$
Convolution	$\int_0^t x_1(\tau)\, x_2(t-\tau)\, d\tau$	$\Longleftrightarrow$	$X_1(s)\, X_2(s)$
Conjugation	$x^*(t)$	$\Longleftrightarrow$	$X^*(s^*)$
Initial value (if it exists)	$x(0^+)$	$\Longleftrightarrow$	$\lim_{s\to\infty} s\, X(s)$
Final value (if it exists)	$x(\infty)$	$\Longleftrightarrow$	$\lim_{s\to 0} s\, X(s)$

Fourier Transform Tables

Specific Fourier Transform Pairs			
Signal	$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df$		$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$
Unit Impulse	$\delta(t)$	$\Longleftrightarrow$	1
Constant	1	$\Longleftrightarrow$	$\delta(f)$
Unit Step	$u(t) = \begin{cases} 1, & t > 0 \\ 0, & t < 0 \end{cases}$	$\Longleftrightarrow$	$\frac{\delta(f)}{2} + \frac{1}{j2\pi f}$
Signum	$\mathrm{sgn}(t) = \begin{cases} 1, & t > 0 \\ 0, & t = 0 \\ -1, & t < 0 \end{cases}$	$\Longleftrightarrow$	$\frac{1}{j\pi f}$
Exponential 1-sided	$e^{-at} u(t)$	$\Longleftrightarrow$	$\frac{1}{a + j2\pi f}$
Gaussian	$e^{-\pi t^2}$	$\Longleftrightarrow$	$e^{-\pi f^2}$
Sine	$\sin(2\pi f_0 t)$	$\Longleftrightarrow$	$\frac{\delta(f - f_0) - \delta(f + f_0)}{2j}$
Cosine	$\cos(2\pi f_0 t)$	$\Longleftrightarrow$	$\frac{\delta(f - f_0) + \delta(f + f_0)}{2}$
Sine $(t > 0)$	$\sin(2\pi f_0 t)\, u(t)$	$\Longleftrightarrow$	$\frac{\delta(f - f_0) - \delta(f + f_0)}{4j} + \frac{1}{2\pi} \frac{f_0}{f_0^2 - f^2}$
Cosine $(t > 0)$	$\cos(2\pi f_0 t)\, u(t)$	$\Longleftrightarrow$	$\frac{\delta(f - f_0) + \delta(f + f_0)}{4} + \frac{1}{j2\pi} \frac{f}{f^2 - f_0^2}$
Rectangular	$x(t) = \begin{cases} 1, & \lvert t \rvert < \tau/2 \\ 0, & \text{otherwise} \end{cases}$	$\Longleftrightarrow$	$\frac{\sin(\pi f \tau)}{\pi f}$
"sin x over x"	$\frac{\sin(2\pi W t)}{\pi t}$	$\Longleftrightarrow$	$X(f) = \begin{cases} 1, & \lvert f \rvert < W \\ 0, & \text{otherwise} \end{cases}$
Hilbert Transform	$\frac{1}{\pi t}$	$\Longleftrightarrow$	$-j\, \mathrm{sgn}(f)$
Tone FM	$e^{j\beta \sin(2\pi f_m t)} = \sum_{k=-\infty}^{\infty} J_k(\beta)\, e^{j2\pi k f_m t}$	$\Longleftrightarrow$	$\sum_{k=-\infty}^{\infty} J_k(\beta)\, \delta(f - k f_m)$
Pulse Train	$\sum_{m=-\infty}^{\infty} \delta(t - mT)$	$\Longleftrightarrow$	$\frac{1}{T} \sum_{k=-\infty}^{\infty} \delta\big(f - \frac{k}{T}\big)$

Bessel function, 1'st kind, order k: $J_k(\beta) = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{j(\beta \sin x - kx)}\, dx$.

Fourier Transform Properties			
Property	$x(t) = \int_{-\infty}^{\infty} X(f) e^{j2\pi ft} df$		$X(f) = \int_{-\infty}^{\infty} x(t) e^{-j2\pi ft} dt$
Linearity	$Ax_1(t) + Bx_2(t)$	$\Longleftrightarrow$	$AX_1(f) + BX_2(f)$
Duality	$X(t)$	$\Longleftrightarrow$	$x(-f)$
Scale Change	$x(at), \quad a \neq 0$	$\Longleftrightarrow$	$\frac{1}{\lvert a\rvert} X\left(\frac{f}{a}\right)$
Time Shift	$x(t - t_0)$	$\Longleftrightarrow$	$X(f)\, e^{-j2\pi f t_0}$
Frequency Shift	$x(t)\, e^{j2\pi f_0 t}$	$\Longleftrightarrow$	$X(f - f_0)$
Convolution	$\int_{-\infty}^{\infty} x(\tau)\, h(t - \tau)\, d\tau$	$\Longleftrightarrow$	$X(f)\, H(f)$
Multiplication	$v(t)\, w(t)$	$\Longleftrightarrow$	$\int_{-\infty}^{\infty} V(\mu)\, W(f - \mu)\, d\mu$
Differentiation	$\frac{dx(t)}{dt}$	$\Longleftrightarrow$	$j2\pi f\, X(f)$
Times t	$t\, x(t)$	$\Longleftrightarrow$	$\frac{-1}{j2\pi} \frac{dX(f)}{df}$
Integration	$\int_{-\infty}^{t} x(\tau)\, d\tau$	$\Longleftrightarrow$	$\frac{X(f)}{j2\pi f} + \frac{X(0)}{2}\, \delta(f)$
Even Symmetry	$x(t) = x(-t)$	$\Longleftrightarrow$	$X(f) = X(-f)$
Odd Symmetry	$x(t) = -x(-t)$	$\Longleftrightarrow$	$X(f) = -X(-f)$
Real $x(t)$	$x(t) = x^*(t)$	$\Longleftrightarrow$	$X(f) = X^*(-f)$ ($\lvert X(f)\rvert$ even, $\angle X(f)$ odd)
Periodic $x(t)$ (X_k: FS Coefficients)	$\sum_{k=-\infty}^{\infty} X_k\, e^{j2\pi kt/T_1}$	$\Longleftrightarrow$	$\sum_{k=-\infty}^{\infty} X_k\, \delta\left(f - \frac{k}{T_1}\right)$
Parseval	$\int_{-\infty}^{\infty} \lvert x(t)\rvert^2\, dt$	$=$	$\int_{-\infty}^{\infty} \lvert X(f)\rvert^2\, df$

Fourier series (period T_1): $X_k = \frac{1}{T_1} \int_{T_1} x(t)\, e^{-j2\pi kt/T_1}\, dt \quad \Longleftrightarrow \quad x(t) = \sum_{k=-\infty}^{\infty} X_k e^{j2\pi kt/T_1}.$

Fourier Series Tables

Specific Fourier Series Pairs, T_1: Period			
Signal	$x(t) = \sum_{k=-\infty}^{\infty} X_k \, e^{j2\pi kt/T_1}$		$X_k = \frac{1}{T_1}\int_{T_1} x(t)\, e^{-j2\pi kt/T_1}\, dt$
Unit Impulse	$\delta(t)$	$\Longleftrightarrow$	$\frac{1}{T_1}$
Constant	1	$\Longleftrightarrow$	$\frac{\sin(\pi k)}{\pi k} = \begin{cases} 1, & k = 0 \\ 0, & \text{otherwise} \end{cases}$
Complex Exponential	$e^{\pm j2\pi mt/T_1}$	$\Longleftrightarrow$	$\frac{\sin(\pi(k \mp m))}{\pi(k \mp m)} = \begin{cases} 1, & k = \pm m \\ 0, & \text{otherwise} \end{cases}$
Sine, $f = \frac{m}{T_1}$	$\sin(2\pi mt/T_1)$	$\Longleftrightarrow$	$\begin{cases} 1/(2j), & k = m \\ -1/(2j), & k = -m \\ 0, & \text{otherwise} \end{cases}$
Cosine, $f = \frac{m}{T_1}$	$\cos(2\pi mt/T_1)$	$\Longleftrightarrow$	$\begin{cases} 1/2, & \lvert k \rvert = m \\ 0, & \text{otherwise} \end{cases}$
Rectangular	$x(t) = \begin{cases} 1, & \lvert t \rvert < \tau/2 \\ 0, & \text{otherwise} \end{cases}$	$\Longleftrightarrow$	$\frac{\sin(\pi k\tau/T_1)}{\pi k}$

Fourier Series Properties, T_1: Period			
Property	$x(t) = \sum_{k=-\infty}^{\infty} X_k \, e^{j2\pi kt/T_1}$		$X_k = \frac{1}{T_1}\int_{T_1} x(t)\, e^{-j2\pi kt/T_1}\, dt$
Linearity	$Ax_1(t) + Bx_2(t)$	$\Longleftrightarrow$	$AX_1[k] + BX_2[k]$
Scale Change	$x(at), \ \left(x(t) = 0,\ \lvert t \rvert \geq \frac{T_1}{a}\right)$	$\Longleftrightarrow$	$\frac{1}{\lvert a \rvert} X[k/a], \quad a \neq 0$
Time Shift	$x(t - t_0)$	$\Longleftrightarrow$	$X_k \, e^{-j2\pi kt_0/T_1}$
Frequency Shift	$x(t)\, e^{j2\pi k_0 t/T_1}$	$\Longleftrightarrow$	X_{k-k_0}
Convolution	$\frac{1}{T_1}\int_{T_1} x(\tau)\, h(t-\tau)\, d\tau$	$\Longleftrightarrow$	$X_k \, H_k$
Multiplication	$v(t)\, w(t)$	$\Longleftrightarrow$	$\sum_{m=-\infty}^{\infty} V_m \, W_{k-m}$
Even Symmetry	$x(t) = x(-t)$	$\Longleftrightarrow$	$X_k = X_{-k}$
Odd Symmetry	$x(t) = -x(-t)$	$\Longleftrightarrow$	$X_k = -X_{-k}$
Real $x(t)$	$x(t) = x^*(t)$	$\Longleftrightarrow$	$X_k = X_{-k}^*$ $(\lvert X_k \rvert$ even, $\angle X_k$ odd$)$
Parseval	$\frac{1}{T_1}\int_{T_1} \lvert x(t) \rvert^2 \, dt$	$=$	$\sum_{k=-\infty}^{\infty} \lvert X_k \rvert^2$

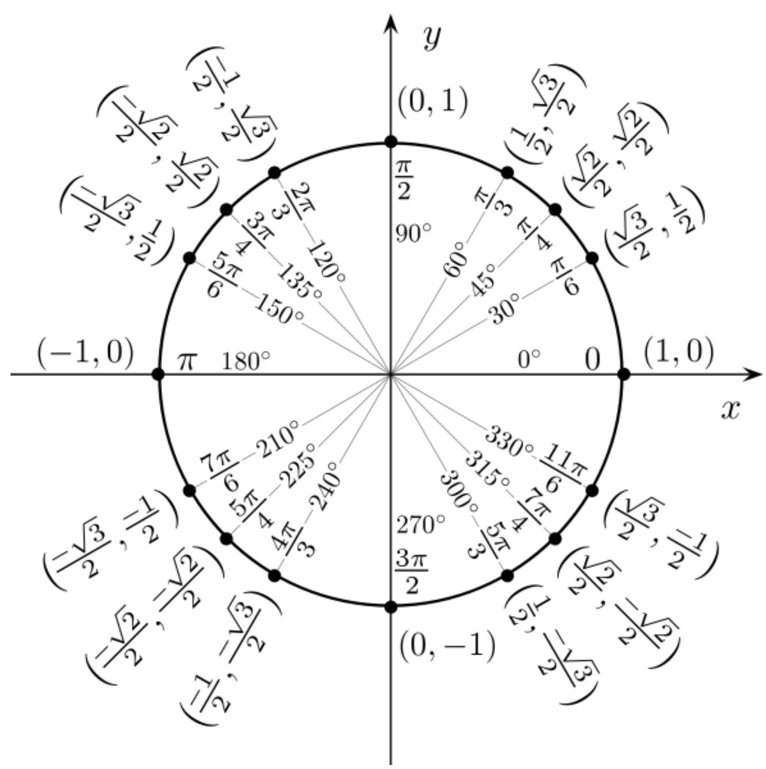
Discrete-Time Fourier Transform Tables

Specific Discrete-Time Fourier Transform Pairs			
Signal	$x_n = \int_1 X(\phi) e^{j2\pi\phi n} d\phi$		$X(\phi) = \sum_{n=-\infty}^{\infty} x_n e^{-j2\pi\phi n}$
Unit Impulse	$\delta_n = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases}$	$\Longleftrightarrow$	1
Constant	1	$\Longleftrightarrow$	$\sum_{k=-\infty}^{\infty} \delta(\phi - k)$
Unit Step	$u_n = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases}$	$\Longleftrightarrow$	$\frac{1}{1 - e^{-j2\pi\phi}} + \frac{1}{2} \sum_{k=-\infty}^{\infty} \delta(\phi - k)$
Signum	$\mathrm{sgn}(n) = \begin{cases} 1, & n > 0 \\ 0, & n = 0 \\ -1, & n < 0 \end{cases}$	$\Longleftrightarrow$	$\frac{1}{j \tan(\pi\phi)}$
Exponential 1-sided	$x_n = \begin{cases} a^n, & n \geq 0,\ \lvert a \rvert < 1 \\ 0, & \text{otherwise} \end{cases}$	$\Longleftrightarrow$	$\frac{1}{1 - a\, e^{-j2\pi\phi}}$
Sine	$\sin(2\pi\phi_0 n)$	$\Longleftrightarrow$	$\sum_{k=-\infty}^{\infty} \frac{\delta(\phi - \phi_0 - k) - \delta(\phi + \phi_0 - k)}{2j}$
Cosine	$\cos(2\pi\phi_0 n)$	$\Longleftrightarrow$	$\sum_{k=-\infty}^{\infty} \frac{\delta(\phi - \phi_0 - k) + \delta(\phi + \phi_0 - k)}{2}$
Rectangular	$x_n = \begin{cases} 1, & \lvert n \rvert \leq d \\ 0, & \text{otherwise} \end{cases}$	$\Longleftrightarrow$	$\frac{\sin\big(\pi\phi(2d+1)\big)}{\sin(\pi\phi)}$
“sin x over x”	$2B \frac{\sin(2\pi B n)}{2\pi B n}$	$\Longleftrightarrow$	$X(\phi) = \begin{cases} 1, & \lvert \phi - k \rvert < B,\ k \text{ integer} \\ 0, & \text{otherwise} \end{cases}$
Hilbert Transform	$x_n = \begin{cases} 2/(\pi n) & n \text{ odd} \\ 0, & n \text{ even} \end{cases}$	$\Longleftrightarrow$	$-j\, \mathrm{sgn}(\phi),\ \frac{-1}{2} < \phi < \frac{1}{2}$
Pulse Train	$\sum_{m=-\infty}^{\infty} \delta_{n-mN}$	$\Longleftrightarrow$	$\frac{1}{N} \sum_{m=-\infty}^{\infty} \delta\big(\phi - \frac{m}{N}\big)$

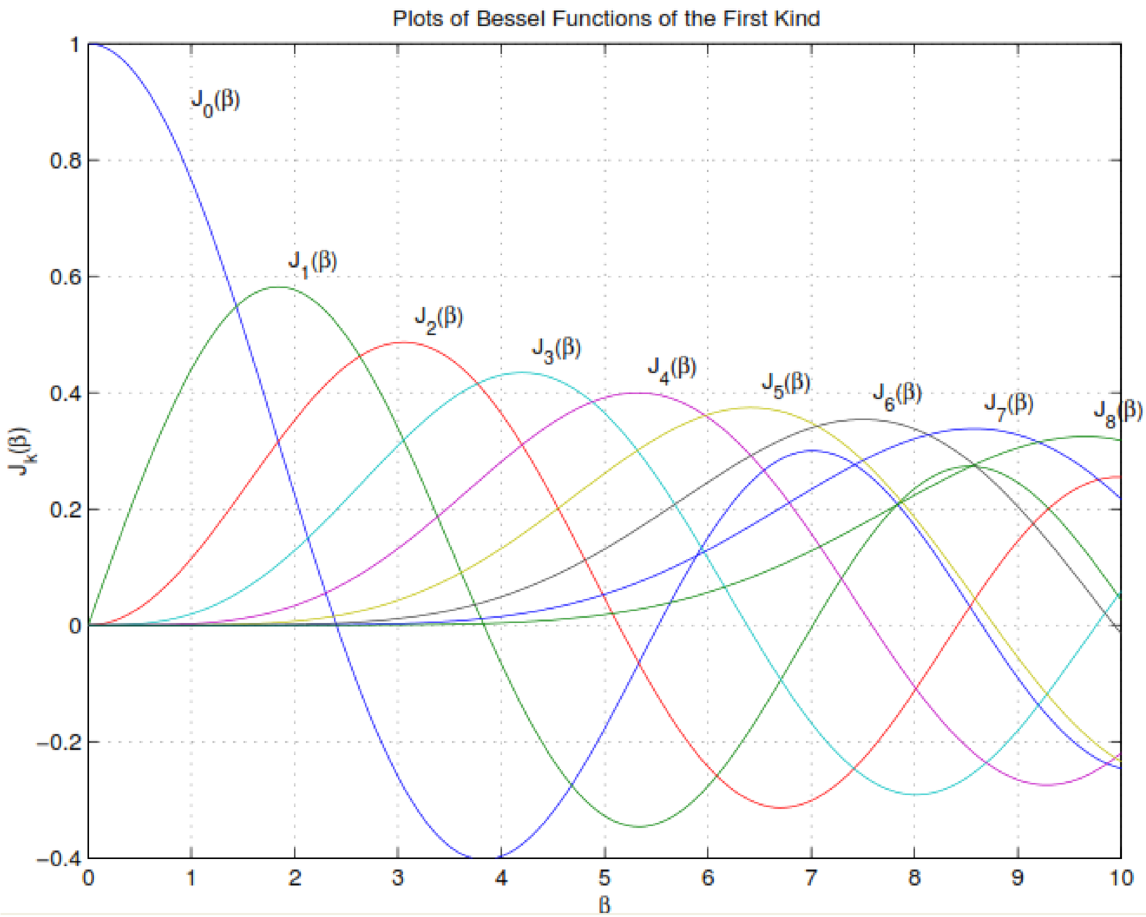
Discrete-Time Fourier Transform Properties			
Property	$x_n = \int_1 X(\phi) e^{j2\pi\phi n} d\phi$		$X(\phi) = \sum_{n=-\infty}^{\infty} x_n e^{-j2\pi\phi n}$
Linearity	$Ax_1[n] + Bx_2[n]$	$\Longleftrightarrow$	$AX_1(\phi) + BX_2(\phi)$
Scale Change (insert $N-1$ zeros)	$\displaystyle\sum_{m=-\infty}^{\infty} x_m\, \delta_{n-mN}$	$\Longleftrightarrow$	$X(N\phi)$
Time Shift	x_{n-n_0}	$\Longleftrightarrow$	$e^{-j2\pi\phi n_0}\, X(\phi)$
Frequency Shift	$e^{j2\pi\phi_0 n}\, x_n$	$\Longleftrightarrow$	$X(\phi - \phi_0)$
Convolution	$\sum_{m=-\infty}^{\infty} x_m\, h_{n-m}$	$\Longleftrightarrow$	$X(\phi)\, H(\phi)$
Multiplication	$v_n\, w_n$	$\Longleftrightarrow$	$\int_1 V(\theta)\, W(\phi-\theta)\, d\theta$
Times n	$n\, x_n$	$\Longleftrightarrow$	$\dfrac{-1}{j2\pi}\, \dfrac{dX(\phi)}{d\phi}$
Summation	$\displaystyle\sum_{m=-\infty}^{n} x_m$	$\Longleftrightarrow$	$\dfrac{X(\phi)}{1-e^{-j2\pi\phi}} + \dfrac{X(0)}{2} \displaystyle\sum_{k=-\infty}^{\infty} \delta(\phi-k)$
Even Symmetry	$x_n = x_{-n}$	$\Longleftrightarrow$	$X(\phi) = X(-\phi)$
Odd Symmetry	$x_n = -x_{-n}$	$\Longleftrightarrow$	$X(\phi) = -X(-\phi)$
Real x_n	$x_n = x_n^*$	$\Longleftrightarrow$	$X(\phi) = X^*(-\phi)$ ($\lvert X(\phi)\rvert$ even, $\angle X(\phi)$ odd)
Periodic x_n	$\dfrac{1}{N} \displaystyle\sum_{k=0}^{N-1} X_k\, e^{j2\pi kn/N}$	$\Longleftrightarrow$	$\dfrac{1}{N} \displaystyle\sum_{k=0}^{N-1} X_k \sum_{m=-\infty}^{\infty} \delta\big(\phi - \dfrac{k}{N} - m\big)$
Parseval	$\sum_{n=-\infty}^{\infty} \lvert x_n\rvert^2$	$=$	$\int_1 \lvert X(\phi)\rvert^2\, d\phi$

Trigonometric Identities

$$e^{\pm j\theta} = \cos\theta \pm j\sin\theta$$
$$\cos\theta = \tfrac{1}{2}(e^{j\theta} + e^{-j\theta}) = \sin(\theta + 90^\circ)$$
$$\sin\theta = \tfrac{1}{2j}(e^{j\theta} - e^{-j\theta}) = \cos(\theta - 90^\circ)$$
$$\sin^2\theta + \cos^2\theta = 1$$
$$\cos^2\theta - \sin^2\theta = \cos 2\theta$$
$$\cos^2\theta = \tfrac{1}{2}(1 + \cos 2\theta)$$
$$\cos^3\theta = \tfrac{1}{4}(3\cos\theta + \cos 3\theta)$$
$$\sin^2\theta = \tfrac{1}{2}(1 - \cos 2\theta)$$
$$\sin^3\theta = \tfrac{1}{4}(3\sin\theta - \sin 3\theta)$$
$$\tan\left(\frac{\alpha}{2}\right) = \sqrt{\frac{1-\cos\alpha}{1+\cos\alpha}} = \frac{\sin\alpha}{1+\cos\alpha} = \frac{1-\cos\alpha}{\sin\alpha}$$
$$\sin(\alpha \pm \beta) = \sin\alpha\,\cos\beta \pm \cos\alpha\,\sin\beta$$
$$\cos(\alpha \pm \beta) = \cos\alpha\,\cos\beta \mp \sin\alpha\,\sin\beta$$
$$\tan(\alpha \pm \beta) = \frac{\tan\alpha \pm \tan\beta}{1 \mp \tan\alpha\,\tan\beta}$$
$$\sin\alpha\,\sin\beta = \tfrac{1}{2}\big[\cos(\alpha-\beta) - \cos(\alpha+\beta)\big]$$
$$\cos\alpha\,\cos\beta = \tfrac{1}{2}\big[\cos(\alpha-\beta) + \cos(\alpha+\beta)\big]$$
$$\sin\alpha\,\cos\beta = \tfrac{1}{2}\big[\sin(\alpha-\beta) + \sin(\alpha+\beta)\big]$$



Bessel Functions



$J_k(\beta) = \dfrac{1}{2\pi}\displaystyle\int_{-\pi}^{\pi} e^{j(\beta \sin x - kx)}\, dx$, Bessel function of 1'st kind, order k.

Properties:

1. $J_k(\beta) = J_k^*(\beta)$
2. $J_{-k}(\beta) = (-1)^k\, J_k(\beta)$
3. $\displaystyle\sum_{k=-\infty}^{\infty} J_k^2(\beta) = 1$

ASCII Code

7-Bit ASCII (American Standard Code for Information Interchange)								
	000...	001...	010...	011...	100...	101...	110...	111...
..0000	*NUL*	*DLE*	*SP*	0	@	P	`	p
..0001	*SOH*	*DC1*	!	1	A	Q	a	q
..0010	*STX*	*DC2*	"	2	B	R	b	r
..0011	*ETX*	*DC3*	#	3	C	S	c	s
..0100	*EOT*	*DC4*	$	4	D	T	d	t
..0101	*ENQ*	*NAK*	%	5	E	U	e	u
..0110	*ACK*	*SYN*	&	6	F	V	f	v
..0111	*BEL*	*ETB*	'	7	G	W	g	w
..1000	*BS*	*CAN*	(	8	H	X	h	x
..1001	*HT*	*EM*	)	9	I	Y	i	y
..1010	*LF*	*SUB*	*	:	J	Z	j	z
..1011	*VT*	*ESC*	+	;	K	[	k	{
..1100	*FF*	*FS*	,	<	L	\	l	|
..1101	*CR*	*GS*	-	=	M	]	m	}
..1110	*SO*	*RS*	.	>	N	^	n	~
..1111	*SI*	*US*	/	?	O	_	o	*DEL*